

Independent Events

Event A does not depend on event B

A, B are said to be independent events when

$$P(A \cap B) = P(A) * P(B)$$

$$P(A|B) = \frac{n(A \cap B)}{n(B)} \quad \text{--- (i)}$$

$$P(A) = \frac{P(A \cap B)}{P(B)} = \frac{n(A \cap B)}{\cancel{n(B)}} * \frac{\cancel{n(B)}}{n(B)}$$

$$P(A) = \frac{n(A \cap B)}{n(B)} \quad \text{--- (ii)}$$

$\therefore$ LHS = RHS

$$\boxed{P(A|B) = P(A)}$$